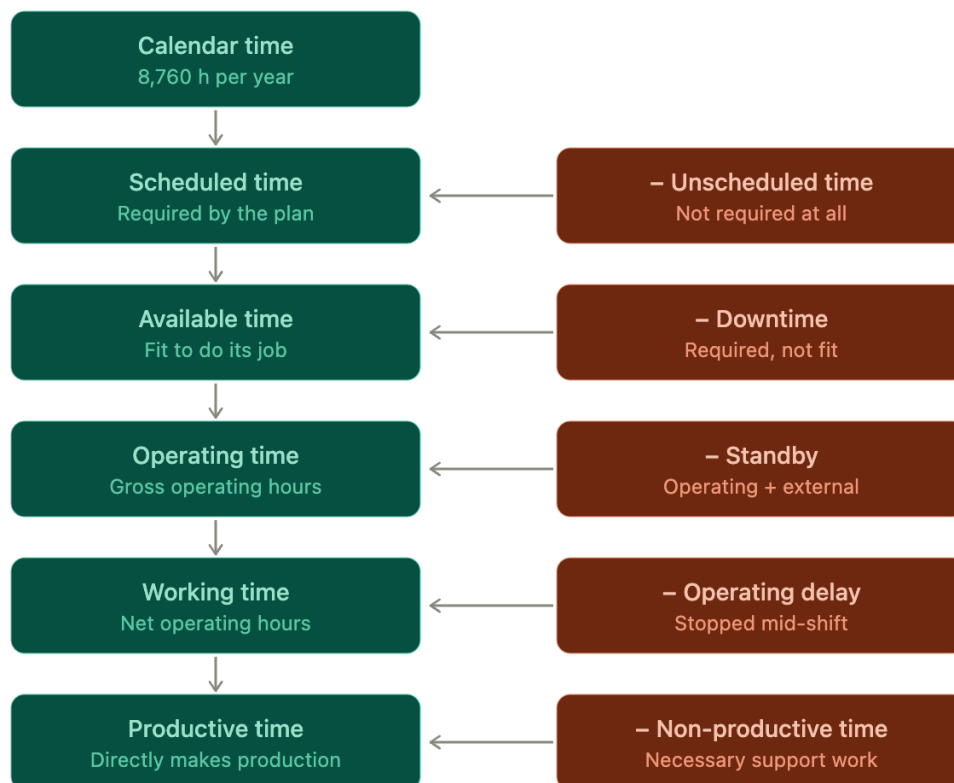


GMG Time Utilisation Model

- Standardises benchmarks for accurate *Mobile Equipment* comparison between organisations

Classification Layers

- Classification layer organises activities, statuses and events into time buckets + allowing for consistent Key Performance Indicators (KPIs)
- Operation must account for all time
- One-to-one 'time-category' linking with Calendar time as root



Calendar Time

- All the time there is in the reporting window. 8,760 h per year, 8,784 in a leap year; the guideline treats the difference as negligible. Also called nominal time
- `timestamp, timestamp_unix`

Scheduled Time

- Time the asset is required to meet the business plan and is assigned to an operation, project or job. $ST = CT - UT$
- Requires Unscheduled Time to be calculated**

Unscheduled Time

- Time the asset isn't assigned at all, because nothing requires it.
- Planned shutdowns, holidays not worked, no work available, mobilisation and demobilisation, major rebuilds.
- Condition of the machine is irrelevant here
- **Requires service hours in dataset**

Available Time

- Required *and* fit for duty. $AT = ST - DT$. About condition, not use — an idle but serviceable machine is still available.
- **Requires Scheduled time and Downtime to be calculated**

Downtime

- The operation wants the asset but it isn't in a condition to perform its function.
- Left as one undivided bucket in the 2020 guideline; no industry consensus on splitting planned from unplanned maintenance, so that detail belongs in a CMMS.
- **Not derivable from datasets as need fault/maintenance log**

Operating Time

- Available but not operating. $SB = SB0 + SBE$
- **Operating standby (SB0)** — No immediate intent to run, for reasons within management control: no operator, shift change, crib breaks, meetings, training, no assignment issued.
- **External standby (SBE)** — Available, required and committed, but blocked by causes outside operating management's influence: client suspends work, work area closed by geotech or water, site-wide weather or power loss, workforce shortage. For a contractor this is the "off hours" state used for billing.
- `timestamp`, `timestamp_unix`, `position` to determine when away from from depot

Standby

- Available and under the control of a human *or a system*. $OT = AT - SB$.
- Conventionally gross operating hours (GOH).
- The "or system" wording is what makes the model work for autonomous fleets.
- `timestamp`, `timestamp_unix`, `position` Partially possible by investigating when stationary at depot during service hours

Working Time

- Running but temporarily stopped by delays inherent to the operation or the immediate physical conditions.
- Within the operator's control conventionally, or the control system's control autonomously.
- Waiting for assignment, stuck, spill cleanup with operator aboard, loss of GPS or site wireless, safety stops.

Operating Delay

- Operating as assigned and performing its intended function, covering activities that do and don't directly produce.
- $WT = OT - OD$
- Conventionally net operating hours (NOH)
- `timestamp`, `timestamp_unix`, `position`, partially possible - unplanned stops

Productive Time

- Unavoidable activity that doesn't directly produce but enables safe, efficient operation to continue.
- Face cleanup, moving trailing cables, tramming, travel empty, waiting at the loading unit.
- `timestamp`, `timestamp_unix`, possible if # of students in bus is counted

Non-Productive Time

- Performing its intended function on activities that directly contribute to production.
- $PT = WT - NP$
- If NP can't be isolated, fold the two together and report at working time.
- `timestamp`, `timestamp_unix`, possible if # of students in bus is counted

KPIs

- A measurable value showing how the business is performing on a key function.

Uptime

- AT / CT
- Time the asset was capable of operating, scheduled or not.
- Alternative offered in the guideline: calendar availability, $(CT - DT) / CT$, which strips effect of unscheduled time dragging down metric (often commercial decision over equipment problem)

Mechanical Availability

- $OT / (OT + DT)$

- The cleanest read on maintenance impact.
- Numerator is operating time, not available time, so standby is excluded and idle hours can't inflate it.
- Contrast physical availability, AT / ST , which does absorb standby.

Use of availability

- OT / AT
- How well the operation uses the equipment it has in working order.
- the maintenance-versus-operations dividing line: high mechanical availability with low use of availability means the fleet is fine and the operation isn't feeding it work.

Effective Utilisation

- WT / ST
- Intended-function time against scheduled time.
- Clears downtime, standby and operating delay in one ratio, making it the closest single number to "did we get value from the shift".

Operating Efficiency

- WT / OT
- Delay burden once the machine is running.
- For an autonomous shuttle this is close to a direct readout of how often it stops for obstacles and pedestrians.

Physical Availability

- AT / ST

Asset Utilisation

- OT / CT

Operating Efficiency

- WT / OT

Production Effectiveness

- PT / OT